

Basic Encryption & Decryption

GitHub README / Project Documentation

■ Project Description

Basic Encryption & Decryption is a beginner-friendly cybersecurity project that demonstrates how plain text can be transformed into encrypted text and then converted back into its original form. The project uses a simple Caesar Cipher technique, where each alphabetic character is shifted by a fixed number of positions.

■ Project Goal

The main goal is to understand the basic idea of encryption and decryption through a simple logic-based implementation. The project helps beginners learn how data can be transformed to make it less readable to unauthorized users and then restored using the correct decryption logic.

■ Key Features

Feature	Description
Text Encryption	Converts user-provided plain text into encrypted text using a Caesar Cipher.
Text Decryption	Converts the encrypted text back into the original plain text.
Custom Shift	Allows a fixed shift value to be used for the encryption/decryption process.
Clear Output	Displays the original, encrypted, and decrypted text.
Beginner Friendly	Uses simple programming concepts such as strings, loops, conditions, and functions.

■ How the Caesar Cipher Works

The Caesar Cipher replaces each letter with another letter a fixed number of positions away in the alphabet. For example, with a shift of 3: A becomes D, B becomes E, and C becomes F. During decryption, the same shift is moved in the opposite direction.

Example:

```
Plain Text: HELLO
Shift: 3
Encrypted Text: KH00R
Decrypted Text: HELLO
```

■ Basic Logic

Encryption: Move each alphabetic character forward by the selected shift value.

Decryption: Move each alphabetic character backward by the same shift value.

Spaces, numbers, and special characters can be kept unchanged depending on the implementation.

■■ Technologies / Skills Used

- Python programming
- String handling
- Loops and conditional statements
- Functions
- Basic encryption concepts
- Basic data protection concepts

■ Suggested GitHub Repository Structure

```
basic-encryption-decryption/  
■■■ app.py           # Main Python program  
■■■ README.md        # Project documentation  
■■■ screenshots/     # Project output screenshots  
■■■ requirements.txt  # Dependencies, if required
```

■■ How to Run

- Install Python on your system.
- Clone or download the GitHub repository.
- Open the project folder in VS Code or another code editor.
- Run the Python file using the command: `python app.py`
- Enter the text and shift value when prompted.
- Check the encrypted and decrypted output.

■ Sample Output

```
Enter text: HELLO WORLD  
Enter shift: 3  
  
Encrypted text: KH00R ZRU0G  
Decrypted text: HELLO WORLD
```

■ Learning Outcomes

- Understand the difference between plaintext, ciphertext, encryption, and decryption.
- Learn how a simple substitution cipher works.
- Practice string manipulation and character processing in Python.
- Understand how the same key/shift can be used in reverse for decryption.
- Gain basic awareness of data protection and cybersecurity concepts.

■■ Security Note

The Caesar Cipher is an educational and historical cipher, not a secure encryption method for protecting real passwords, confidential files, or sensitive information. Modern applications should use well-tested cryptographic algorithms and libraries.

■ Future Improvements

- Add a graphical web interface using HTML/CSS and Python.
- Add support for multiple encryption methods.
- Add input validation and error handling.
- Allow users to select the encryption shift.
- Add a copy-to-clipboard feature.
- Add a modern UI with encryption/decryption status messages.

■ GitHub Short Description

A beginner-friendly Python cybersecurity project that implements Caesar Cipher encryption and decryption to demonstrate basic data protection concepts.

■■ Suggested GitHub Topics

python, encryption, decryption, caesar-cipher, cybersecurity, cryptography, beginner-project, string-handling, information-security